Esercizio 1

S_n = numero di teste uscite nei primi n lanci

$S_n \sim Bin(n, p)$ $\quad n = 1, \dots, 100$ $\quad p = 1/4$

a)

$$\mathbb{P}[S_5 \geq 4] = \mathbb{P}[S_5 = 4] + \mathbb{P}[S_5 = 5] = \binom{5}{4}\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)^1 + \binom{5}{5}\left(\frac{1}{4}\right)^5$$

$$= \frac{15}{4^5} + \frac{1}{4^5} = \frac{16}{4^5} = \frac{1}{64} \simeq 1.56\,\%$$

b) $$\mathbb{P}[S_5 = 4 \mid S_2 = 2] = \mathbb{P}[(S_5 - S_2) + S_2 = 4 \mid S_2 = 2] =$$

$$= \mathbb{P}[S_5 - S_2 = 2] = \mathbb{P}[S_3 = 2] = \binom{3}{2}\left(\frac{1}{4}\right)^2 \frac{3}{4}$$

$(S_5 - S_2) \perp S_2$ $\qquad S_5 - S_2 \sim S_3$

$$= 3 \cdot \frac{3}{4^3} = \frac{9}{64}$$

c) $$\mathbb{E}[S_{100}] = n p = 100 \cdot \frac{1}{4} = 25$$

$$VAR[S_{100}] = n p(1-p) = 100 \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{75}{4}$$

d) $$\mathbb{P}[S_{100} \leq 40] = \mathbb{P}\left[\frac{S_n - \mathbb{E}[S_{100}]}{\sqrt{VAR[S_{100}]}} \leq \frac{35 - 25}{\sqrt{75/4}}\right]$$

$$\simeq \mathbb{P}[Z \leq 2.31] \simeq 98.85$$

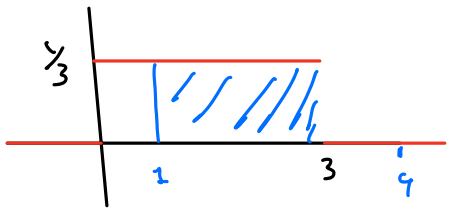
$Z \sim N(0, 1)$

Esercizio 2

$X \sim \mathcal{U}[0, 3]$ $\quad W \sim B(1/3)$ $\quad Y = X^3$

a)

$$f(x) = \begin{cases} \frac{1}{3} & 0 \le x \le 3 \\ 0 & \text{altrove} \end{cases}$$



$$\mathbb{E}[X] = \frac{a+b}{2} = \frac{3}{2} \qquad \mathrm{VAR}[X] = \frac{(b-a)^2}{12} = \frac{9}{12} = \frac{3}{4}$$

$a = 0 \quad b = 3$

b) $\mathbb{P}[1 \le X \le 4] = \frac{1}{3}(3-1) = \frac{2}{3} = 66.7\%$

$\mathbb{P}[X = 1 \text{ o } X = 2] = 0$ poiché uniche con densità

c) $\mathbb{E}[e^{XW}] = \mathbb{P}[\underbrace{e^0}_{1} \mid W=0]\,\mathbb{P}[W=0] + \mathbb{E}[e^X \mid W=1]\,\mathbb{P}[W=1]$

$$= 1 \cdot \frac{1}{5} + \frac{4}{5}\int_0^3 e^x \frac{1}{3}\,dx = \frac{1}{5} + e^x\Big|_0^3 \frac{4}{15}$$

$$= \frac{3 + 4(e^3 - 1)}{15} \approx 5.39$$

d) $\mathbb{E}[Y] = \mathbb{E}\left[\frac{X^3}{27}\right] = \frac{1}{27}\cdot\frac{1}{3}\int_0^3 x^3\,dx = \frac{1}{81}\cdot\frac{x^4}{4}\Big|_0^3 = \frac{1}{4}$

e) $0 \le X \le 3 \qquad 0 \le Y \le \frac{3^3}{27} = 1$

se $0 < y < 1$

$$g(y) = \begin{cases} 0 & \text{se } y < 0 \\ 0 & \text{se } y \ge 1 \end{cases}$$

$$G(y) = \mathbb{P}[Y \le y] = \mathbb{P}\left[\frac{X^3}{27} \le y\right] = \mathbb{P}[X \le 3\sqrt[3]{y}]$$

$$= \sqrt[3]{y}$$

$$g(y) = G'(y) = \left(3\sqrt[3]{y}\right)' = \frac{1}{3}\, y^{-2/3} = \frac{1}{3}\,\frac{1}{\sqrt[3]{y^2}}$$

$$g(y) = \begin{cases} 0 & y \le 0 \\ \frac{1}{3}\,\frac{1}{\sqrt[3]{y^2}} & 0 < y < 1 \\ 0 & y \ge 1 \end{cases}$$

Ejercicio 3

a) $t_{0.005,50} = 2.68$

$\chi^2_{0.005,50} = 29.49$ $\qquad$ $\chi^2_{0.995,50} = 27.99$

b)

$I = [\bar{x} - s,\ \bar{x} + s]$ $\qquad$ $s = \dfrac{t_{0.005,50}\ s}{\sqrt{n}} \simeq 3.75$

$I = [143.25,\ 150.75]$

c) $I = \left[\dfrac{s^2}{\frac{\chi^2_{0.005,50}}{n-1}},\ \dfrac{s^2}{\frac{\chi^2_{0.995,50}}{n-1}} \right] = [62.9,\ 178.6]$

d)

$H_0 : \mu = \mu_0 \quad \cdot \quad \mu_0 = 150\ g$

$H_1 : \mu \neq \mu_0$

$$\hat{t} = \frac{|\bar{x} - \mu_0|}{s/\sqrt{n}} \simeq 2.12124$$

α	$t_{n-1,\alpha}$		
1%	2.6778	accetto	H_0
5%	2.008	rifiuto	H_0